What Do Neural Predictors Add to Budget Pacing?

Spend Forecasts, Budget Delivery and Hedging on Criteo Campaign Logs

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Abstract

A platform can pace a daily advertising budget by forecasting the cost of the remaining impressions and choosing what share to keep. Does a neural forecast improve this decision? On 414 Criteo campaigns and a fixed impression-replay throttle, we compare five neural predictors (TFT, NHITS, PatchTST, LSTM, TiDE) with seasonal-naive and LightGBM forecasts at three budget levels. All five capture 0.98–2.14 more percentage points of the hindsight-reference click count than the naive plan, mostly by spending 1.2–2.1% more of the budget. A validation-chosen scale on the naive forecast recovers 36–94% of TFT's gain, and a total–profile swap attributes 68–82% of it to the forecast daily total. Planning with 90th-percentile hourly forecasts costs 7.7–10.6 points; calibration recovers only part of that loss. The evidence covers 5 test days of one dataset, and a small iPinYou check does not reproduce the policy ordering.

CCS Concepts

• **Information systems** → **Online advertising**; • **Computing methodologies** → *Neural networks*; • **Mathematics of computing** → *Time series analysis.*

Keywords

budget pacing, throttling, real-time bidding, probabilistic forecasting, neural predictors

1 Introduction

An advertiser's daily budget may run out before the day ends, or remain partly unspent. Budget pacing controls how quickly the platform spends it [1, 12]. In *probabilistic throttling*, the platform joins each eligible auction with a probability α_t that it updates during the day [1, 18]. A reactive controller, such as a proportional-integral-derivative (PID) controller, adjusts this probability from past spend. A forecast-based rule instead divides the remaining budget by the forecast cost of the remaining impressions. Forecasting too much spend lowers participation and may leave budget unused.

We ask whether neural predictors improve this rule, and what they add. A spend forecast supplies a daily total and an hourly profile; a better total could help even if the profile adds little. We also test whether planning with an upper quantile protects against early exhaustion at an acceptable cost.

Using the public Criteo Attribution dataset [7], we compare five neural predictors with seasonal naive and LightGBM under one fixed throttling rule. We separate extra clicks from extra spend, test a scaled naive baseline, swap totals and profiles, and measure the cost of hedging. This is an empirical comparison of established models, not a new architecture. Follow-up experiments were designed after the main results were known; their decision rules are logged in the accompanying protocol, which reports every analysis in full.

Related work. Smart pacing [18] learns participation rates per segment; Dai et al. [5] control throttling with dual multipliers; Balseiro and Gur [2] adjust bids by dual gradients; and Zhang et al. [19] use feedback toward spend targets. Chen's guide [4] notes that a PID controller cannot repair a misspecified target by tracking it. Shekhar and Howard [16] calibrate forecast uncertainty for pacing decisions and find that the lowest forecast error need not give the best decision. We instead measure how much changing the forecast changes clicks under a fixed rule.

2 Setup

Data and episodes. The Criteo dataset has 16,468,027 impressions from 675 campaigns. We use 21 training, 4 validation and 5 test days, and keep 414 campaigns with spend in at least 90% of training hours (91.7% of dataset spend). The forecast target is *spend demand*: the logged cost of all impressions in a campaign-hour, in Criteo's transformed cost units. An episode is a campaign-day with budget $B = \rho \bar{D}_c$, where $\bar{D}_c$ is the campaign's mean daily training spend and $\rho \in \{0.3, 0.5, 0.7\}$ gives tight, medium and loose budgets.

Replay. At the start of hour h, a policy sets $\alpha_h \in [0, 1]$; each logged impression is kept independently with that probability and adds its logged cost and click, in order, until the budget is spent. All policies share the same random draws (10 seeds per episode). This replays *won* impressions: we assume that keeping fewer of them does not change their prices, and we model no competitor response.

Policies. With S_h the spend so far and $R_h = B - S_h$: *no pacing* keeps everything; *even pacing* steers toward $R_h/(24-h)$ per hour; *PID* tracks the gap to a linear spending schedule, with gains tuned on validation days per budget. The *forecast throttle* sets

$$\alpha_h = \min\Big(1,\ R_h \,/\, \textstyle\sum_{j=h}^{23} \hat{y}_j^{(q)}\Big), \qquad q \in \{0.5, 0.7, 0.9\}, \tag{1}$$

re-planned every hour from hourly quantile forecasts $\hat{y}_j^{(q)}$. A higher q hedges against early exhaustion. The *hindsight constant* $\alpha = \min(1, B/D)$, with D the day's logged spend demand, defines the reference click count; it is not an upper bound.

Forecasters. Each model forecasts the next 24 hourly spends at quantiles $\{0.1, 0.5, 0.7, 0.9\}$ from data before the issue hour. *Seasonal naive* takes quantiles of the same hour on the previous seven days. *LightGBM* [11] fits one quantile model per level on lags, rolling statistics and calendar features. TFT [13], NHITS [3], PatchTST [14], LSTM [8] and TiDE [6] use their `neuralforecast` implementations [15], 168 hours of input and multi-quantile loss, with 3 training seeds each. PatchTST uses only past spend; the others also use impressions, clicks, calendar phase and campaign statistics. Models are early-stopped on validation days and not refitted during testing.

Metrics. The primary metric is total clicks divided by the hindsight reference's clicks. A *point* is one percentage point of this ratio. We also report budget utilization (BUR), early depletion before hour 20, and relative changes in spend and in clicks per unit of spend. Intervals are 95% campaign-bootstrap CIs (2,000 resamples) on paired differences; they condition on the trained models and exclude training-seed uncertainty. Forecast accuracy is MASE [10] scaled by each campaign's training lag-24 error.

3 Results

3.1 Forecast accuracy

Mean MASE is lowest for TFT (0.787), then NHITS (0.794), LightGBM (0.853) and seasonal naive (0.917) (Table 2). The ranking does not hold for every campaign: TFT beats LightGBM in 70.5% of campaigns. All prediction intervals under-cover; the nominal 80% interval covers 76.9% of outcomes for TFT and 60.6% for seasonal naive. Neural times use one RTX 3070 Ti and LightGBM a CPU, so they are not hardware-matched.

3.2 Neural forecasts beat the naive plan

Median-plan TFT and NHITS capture 3.4–5.4 points more clicks than even pacing and 0.7–2.0 more than tuned PID (Table 1); all 12 of 12 such comparisons are significant under Wilcoxon tests with Holm correction [9, 17]. They also change their rates less sharply than PID and deplete early less often (3.4–6.1% versus 12.8–31.9% of episodes).

Table 3 extends the comparison to all five neural models. Each gains 0.98–2.14 points over seasonal naive at every budget, with every CI above zero, meeting the protocol's pre-set rule (four of five models at two of three budgets). Ranking the models by MASE gives the reverse of their ranking by mean click gain (Spearman −1.00), though five models cannot establish a general relationship. PatchTST is competitive without covariates. TFT and NHITS differ by at most 0.2 points, within NHITS's seed SD of up to 0.45.

3.3 Where the gain comes from

More spend, similar efficiency. The neural models spend 1.2–2.1% more than seasonal naive, while clicks per unit of spend change by −0.15% to 0.24%. For TFT, the efficiency CI includes zero at 3 of 3 budget levels (Table 4). Against PID, by contrast, TFT's clicks per spend rise by 1.1–2.1%.

A scaled naive baseline. Multiplying the naive denominator by a scale chosen on validation days (0.85, 0.825, 0.775) recovers 36%, 60% and 94% of TFT's gain at the tight, medium and loose budgets. TFT keeps a clear edge at the two tighter budgets; at the loose budget the difference is 0.08 points (CI [−0.19, 0.37]). Neural models should therefore be compared with a scaled baseline, not only an unscaled one.

Total versus profile. We build fixed plans from day-start forecasts and replace either the naive daily total or its hourly shares with TFT's (Table 5). The total alone captures 68–82% of the combined gain; the profile adds less, though its CI is above zero at the medium and loose budgets. Hourly re-planning adds 0.25–0.51 points over the fixed plan. Replacing TFT's forecast with the actual remaining spend demand adds 1.03–1.54 points: moderate headroom for this rule, not an upper bound on clicks.

Matched spend (exploratory). Interpolating the 25-scale naive curve at each model's total spend leaves all five neural models 0.26–1.30 points above it (Table 3). One global rescaling thus does not reproduce them, but this analysis cannot say whether the remainder comes from campaign-specific totals, profiles or re-planning.

3.4 Hedging

Clicks fall as q rises from 0.5 to 0.9 in all 12 of 12 forecaster-and-budget combinations, costing 7.7–10.6 points (Figure 1). Upper plans cut early depletion to 0.1–0.7% of episodes, but budget utilization falls to 75.2–82.8%. Summed hourly 90% quantiles over-cover the remaining-day total (94.0–97.4% coverage). Rescaling the median plan by factors calibrated on earlier test days recovers 20%, 14% and 11% of TFT's $q = 0.9$ loss, which falls short of the protocol's one-third threshold. The replay rewards clicks and barely penalizes early exhaustion, so this cost partly reflects the objective.

3.5 Robustness checks

Other reactive rules. A tuned dual-gradient throttle [2] does not beat PID, and TFT leads its corrected variant by 2.9–3.6 points. **Feedback.** Adding PID feedback to the forecast rate, with gains tuned on seasonal naive, does not improve TFT at two of three budgets, and TFT then no longer clearly beats scaled naive with feedback. **A synthetic world.** In generated worlds with planted rhythms, replaying the previous week gains 1.5–2.7 points over reactive control, and a perfect daily total adds only 0.1–0.3. **iPinYou.** On 5 advertisers and 10 test episodes, the naive-over-even differences have CIs that include zero, and the Criteo ordering holds at 0 of 3 budget levels. No neural models were tested there.

4 Discussion and Conclusion

On Criteo, all five tested neural predictors capture more clicks than seasonal naive under the same throttle, mainly by spending more of the budget with similar clicks per unit of spend. In TFT's swap experiment most of the gain comes from the daily total. A single validation-chosen scale recovers much of it at the loose budget but less than half at the tight one, and a gain remains at matched aggregate spend whose cause is unresolved. Upper-quantile plans leave budget unspent in this click-focused replay, and calibration recovers only part of that loss.

Limitations. The test period is 5 days of one dataset. Intervals condition on saved models and exclude seed uncertainty. The replay uses won impressions and transformed prices, without price responses or an explicit cost of early exhaustion. Day boundaries are relative to the dataset start. The main design was informed by a full-data profile, and the follow-up analyses reused the same

Table 1: Click ratios of the original four-forecaster study, pooled over episodes and seeds; bold marks the best non-reference value. Operational metrics at $\rho = 0.5$.

	Clicks / reference			At $\rho = 0.5$			
Policy	ρ=0.3	ρ=0.5	ρ=0.7	Conv./ref.	BUR (%)	Early depl. (%)	PVI
Hindsight constant (ref.)	1.000	1.000	1.000	1.000	90.6	9.5	0.000
No pacing	0.985	0.979	0.988	1.018	94.4	80.1	0.000
Even pacing	0.993	0.960	0.942	0.966	90.2	6.4	0.314
PID (tuned)	1.009	0.990	0.988	1.004	92.2	31.9	0.577
Seasonal naive, q=0.5	1.007	0.989	0.983	0.988	91.3	6.4	0.083
LightGBM, q=0.5	1.019	0.997	0.992	0.998	92.4	5.0	0.094
NHITS, q=0.5	**1.028**	**1.007**	0.995	1.005	92.5	4.3	0.118
TFT, q=0.5	1.027	1.007	**0.996**	1.006	92.7	5.5	0.097
Seasonal naive, q=0.7	0.984	0.965	0.959	0.957	87.7	2.4	0.075
LightGBM, q=0.7	0.995	0.972	0.965	0.964	88.7	1.6	0.082
NHITS, q=0.7	1.011	0.985	0.971	0.976	89.1	1.1	0.101
TFT, q=0.7	1.010	0.985	0.974	0.977	89.4	1.6	0.085
Seasonal naive, q=0.9	0.930	0.910	0.904	0.897	79.6	0.6	0.070
LightGBM, q=0.9	0.914	0.892	0.887	0.879	78.2	0.5	0.071
NHITS, q=0.9	0.950	0.919	0.904	0.904	79.6	0.2	0.087
TFT, q=0.9	0.945	0.916	0.904	0.905	79.6	0.3	0.074

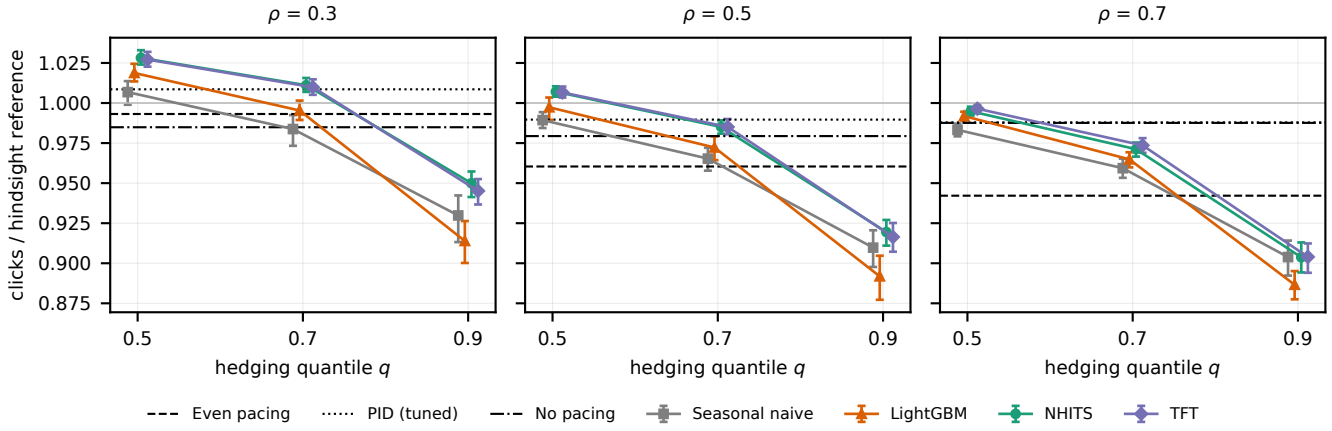


Figure 1: Click ratio versus planning quantile (95% campaign-bootstrap CIs). Horizontal lines are forecast-free policies.

Table 2: Original four forecasters: mean test accuracy (± training-seed SD). MASE_h is MASE at step h; $\text{sPL}_{0.9}$ is scaled pinball loss; Cov. is 80% interval coverage; time includes fitting and forecasting.

Forecaster	MASE	$\text{MASE}_{h=1}$	$\text{MASE}_{h=24}$	$\text{sPL}_{0.9}$	Cov. 80%	Time (s)
Seasonal naive	0.917	0.920	0.908	0.194	60.6	0
LightGBM	0.853	0.810	0.871	0.185	71.7	76
NHITS	0.794 ± 0.014	0.719	0.813	0.158	75.8	40
TFT	0.787 ± 0.004	0.729	0.814	0.161	76.9	205

Table 3: Five neural predictors. MASE averages campaigns and seeds (naive 0.917, LightGBM 0.853). Gains and distances above the spend-matched naive scaling curve are in points.

		Gain over seasonal naive			Above the scaling curve		
Model	MASE	ρ=0.3	ρ=0.5	ρ=0.7	ρ=0.3	ρ=0.5	ρ=0.7
TFT	0.787	2.03	1.73	1.32	1.20	0.74	0.47
NHITS	0.794	2.14	1.77	1.16	1.30	0.83	0.44
PatchTST	0.806	1.82	1.68	1.26	1.00	0.67	0.39
LSTM	0.816	1.60	1.27	0.98	0.81	0.43	0.26
TiDE	0.786	2.05	1.92	1.50	1.21	0.82	0.50

test days after the main results were known. Tuning used only 4 validation days. The iPinYou check is small and tests no neural models. Next steps are seed-aware inference, longer out-of-time evaluation on other real logs, and a simulator with price responses and delivery constraints. Code, protocol and results will be released.

Table 4: Median-plan comparisons. Click gains in points with 95% CIs; spend and clicks per spend are relative changes.

Comparison	ρ	Clicks (points)	Spend (%)	Clicks per spend (%)
TFT − seasonal naive	0.3	2.03 [1.57, 2.62]	1.9	0.15
	0.5	1.73 [1.36, 2.11]	1.8	−0.03
	0.7	1.32 [1.06, 1.63]	1.4	−0.03
Scaled naive − seasonal naive	0.3	0.73 [0.55, 0.93]	1.4	−0.69
	0.5	1.03 [0.82, 1.25]	1.9	−0.83
	0.7	1.24 [0.96, 1.53]	2.4	−1.12
TFT − scaled naive	0.3	1.30 [0.93, 1.77]	0.4	0.84
	0.5	0.69 [0.39, 1.03]	−0.1	0.81
	0.7	0.08 [−0.19, 0.37]	−1.0	1.11

Table 5: Day-start plan swaps and the oracle, in points (95% CIs).

Comparison	$\rho = 0.3$	$\rho = 0.5$	$\rho = 0.7$
Fixed plan − fixed naive plan			
TFT total + TFT profile	1.52	1.35	1.07
TFT total + naive profile	1.25 [0.82, 1.83]	1.02 [0.71, 1.38]	0.72 [0.50, 0.95]
Naive total + TFT profile	0.16 [−0.02, 0.31]	0.17 [0.05, 0.29]	0.25 [0.14, 0.37]
Oracle remaining − TFT	1.03 [0.73, 1.35]	1.50 [1.23, 1.78]	1.54 [1.27, 1.81]

Use of Generative AI Tools

Claude (Anthropic) and an OpenAI assistant helped with code, experiments, writing and draft review. Numerical results come from the accompanying code. Model feedback is not human peer review. The authors are responsible for the content.

References

[1] Deepak Agarwal, Souvik Ghosh, Kai Wei, and Siyu You. 2014. Budget Pacing for Targeted Online Advertisements at LinkedIn. In *Proceedings of the 20th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*. 1613–1619. doi:10.1145/2623330.2623366

[2] Santiago R. Balseiro and Yonatan Gur. 2019. Learning in Repeated Auctions with Budgets: Regret Minimization and Equilibrium. *Management Science* 65, 9 (2019), 3952–3968. doi:10.1287/mnsc.2018.3174

[3] Cristian Challu, Kin G. Olivares, Boris N. Oreshkin, Federico Garza Ramirez, Max Mergenthaler-Canseco, and Artur Dubrawski. 2023. NHITS: Neural Hierarchical Interpolation for Time Series Forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 37. 6989–6997. doi:10.1609/aaai.v37i6.25854

[4] Yuanlong Chen. 2025. A Practical Guide to Budget Pacing Algorithms in Digital Advertising. arXiv:2503.06942.

[5] Liang Dai, Kejie Lyu, Chengcheng Zhang, Guangming Zhao, Zhonglin Zu, Liang Wang, and Bo Zheng. 2024. Percentile Risk-Constrained Budget Pacing for Guaranteed Display Advertising in Online Optimization. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 7987–7994. doi:10.1609/aaai.v38i8.28636

[6] Abhimanyu Das, Weihao Kong, Andrew Leach, Shaan Mathur, Rajat Sen, and Rose Yu. 2023. Long-term Forecasting with TiDE: Time-series Dense Encoder. *Transactions on Machine Learning Research* (2023). arXiv:2304.08424.

[7] Eustache Diemert, Julien Meynet, Pierre Galland, and Damien Lefortier. 2017. Attribution Modeling Increases Efficiency of Bidding in Display Advertising. In *Proceedings of the AdKDD and TargetAd Workshop, KDD*. doi:10.1145/3124749.3124752

[8] Sepp Hochreiter and Jürgen Schmidhuber. 1997. Long Short-Term Memory. *Neural Computation* 9, 8 (1997), 1735–1780. doi:10.1162/neco.1997.9.8.1735

[9] Sture Holm. 1979. A Simple Sequentially Rejective Multiple Test Procedure. *Scandinavian Journal of Statistics* 6, 2 (1979), 65–70. doi:10.2307/4615733

[10] Rob J. Hyndman and Anne B. Koehler. 2006. Another Look at Measures of Forecast Accuracy. *International Journal of Forecasting* 22, 4 (2006), 679–688. doi:10.1016/j.ijforecast.2006.03.001

[11] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. 2017. LightGBM: A Highly Efficient Gradient Boosting Decision Tree. In *Advances in Neural Information Processing Systems (NIPS)*, Vol. 30. 3146–3154.

[12] Kuang-Chih Lee, Ali Jalali, and Ali Dasdan. 2013. Real Time Bid Optimization with Smooth Budget Delivery in Online Advertising. In *Proceedings of the Seventh International Workshop on Data Mining for Online Advertising (ADKDD)*. 1–9. doi:10.1145/2501040.2501979

[13] Bryan Lim, Sercan Ö. Arık, Nicolas Loeff, and Tomas Pfister. 2021. Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting. *International Journal of Forecasting* 37, 4 (2021), 1748–1764. doi:10.1016/j.ijforecast.2021.03.012

[14] Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. 2023. A Time Series is Worth 64 Words: Long-term Forecasting with Transformers. In *International Conference on Learning Representations (ICLR)*. arXiv:2211.14730.

[15] Kin G. Olivares, Cristian Challú, Federico Garza, Max Mergenthaler Canseco, and Artur Dubrawski. 2022. NeuralForecast: User Friendly State-of-the-Art Neural Forecasting Models. PyCon, Salt Lake City, Utah, US. https://github.com/Nixtla/neuralforecast.

[16] Prashant Shekhar and Caroline Howard. 2026. Decision-Calibrated Conformal Uncertainty for Pacing Decisions in Streaming Advertising. arXiv:2606.10187.

[17] Frank Wilcoxon. 1945. Individual Comparisons by Ranking Methods. *Biometrics Bulletin* 1, 6 (1945), 80–83. doi:10.2307/3001968

[18] Jian Xu, Kuang-Chih Lee, Wentong Li, Hang Qi, and Quan Lu. 2015. Smart Pacing for Effective Online Ad Campaign Optimization. In *Proceedings of the 21st ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*. 2217–2226. doi:10.1145/2783258.2788615

[19] Weinan Zhang, Yifei Rong, Jun Wang, Tianchi Zhu, and Xiaofan Wang. 2016. Feedback Control of Real-Time Display Advertising. In *Proceedings of the ACM International Conference on Web Search and Data Mining (WSDM)*. 277–286. doi:10.1145/2835776.2835843